Peer Review Filter Classroom Run Kit

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	**Time**	30-40 min
Difficulty	Medium	**ID**	peer-review-filter

Big idea: Scientific knowledge is filtered by community criticism, not just produced by individual discovery.

One-Minute Teacher Brief

Students act as reviewers for a fictional study and decide whether flaws require revision, rejection, or cautious acceptance.

Student Prompt

Inspect Peer Review Filter Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Peer Review Filter Stimulus A	Student-facing stimulus 1: Study claim: a lavender scent improves memory scores by 18% in one class of 18 students. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Peer Review Filter Stimulus B	Student-facing stimulus 2: Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Peer Review Filter Reveal / Twist	Student-facing stimulus 3: Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Scientific knowledge is filtered by community criticism, not just produced by individual discovery.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Peer Review Filter Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Peer Review Filter Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How does criticism strengthen knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how scientific knowledge is filtered by community criticism, not just produced by individual discovery.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

• Work silently for the first response so you can notice your own starting point.

• Record one knowledge claim, the evidence behind it, and your confidence level.

• After the reveal or comparison, update your claim in a different colour or column.

• Use the debrief to answer: How does criticism strengthen knowledge?

Setup Checklist

• Write the fictional study so it has strengths as well as flaws.

• Give reviewers different priorities to simulate real disagreement.

Example Stimuli or Materials

• Study claim: a lavender scent improves memory scores by 18% in one class of 18 students.

• Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up.

Resources To Use

• Resource pack: resources/peer-review-filter/index.html

• Card deck CSV: resources/peer-review-filter/cards.csv

• Visual prompt: resources/peer-review-filter/visual-prompt.svg

• Reviewer checklist: sample, control, measurement, claim size, alternative explanation, ethical issue.

• Decision card: accept, minor revision, major revision, reject, or invite replication.

Run Sequence

1. Give groups a fictional study with several strengths and weaknesses.
2. Students annotate method, sample, measurement, claim size, and limitations.
3. Groups write reviewer comments and a decision: accept, revise, or reject.
4. Compare reviewer decisions and identify criteria behind disagreement.
5. Debrief peer review as a social process for improving knowledge.

Debrief Moves

- Knowledge question: How does criticism strengthen knowledge?
- Does peer review guarantee truth or reduce risk?
- What should a scientific community do with promising but flawed evidence?

Teacher Quick Script

- Peer review is not a truth machine; it is organized criticism.
- Which flaw changes the conclusion, and which flaw can be repaired?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...